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The role of the market subsystem and the socio-technical subsystem in innovation and early internationalization: A dynamic capabilities approach

Abstract

Organizational learning has been studied as a key factor in firm performance and internationalization. Moving beyond the past emphasis on market learning, we develop a more complete explanation of learning, its relationship to innovation, and their joint effect on early internationalization. We theorize that, driven by the founders' international vision, early internationalizing firms employ a dual subsystem of dynamic capabilities: a market subsystem consisting of market-focused learning capability and marketing capability, and a socio-technical subsystem comprised of network learning capability and internally focused learning capability. We argue that innovation mediates the proposed relationship between the dynamic capability structure and early internationalization. We conduct case studies to develop the conceptual framework and test it in a field survey of early internationalizing firms from Australia and the United States. Our findings indicate a complex interplay of capabilities driving innovation and early internationalization. We provide theoretical and practical implications and offer insights for future research.

Keywords International vision, organizational learning, marketing capability, organizational subsystems, early internationalization, dynamic capabilities, born globals, innovation

Firms develop knowledge through the acquisition, dissemination, interpretation, and storage of information from their environments (Hanvanich et al. 2006; Huber 1991). The role of organizational learning in knowledge development and firm performance has received substantial attention in both the strategic marketing and internationalization literatures. While marketing scholars recognize the importance of organizational learning (Peters et al. 2009), understanding of its nature and role remains relatively limited (e.g., Friedman et al. 2005). Scholars have focused primarily on market learning in firm performance, which is represented by the market-driven (Day 1994 a, b; Sinkula et al. 1997; Slater and Narver 1995) and market orientation (Kohli and Jaworski 1990) perspectives. This focus reveals a gap in the broader literature on organizational learning.

In addition to market knowledge, firms seek new knowledge through networks (Selnes and Sallis 2003) and internal experimentation (Cohen and Levinthal 1990). These internally focused and externally focused learning activities represent different types of knowledge that are not substitutes for one another but are complementary (Arora and Gambardella 1990; Powell et al. 1996). Collectively they have been observed as important precursors to innovation (Hartman et al. 1994), firm performance (Bierly and Chakrabarti 1996), and early internationalization (Oviatt and McDougall 2005). How these learning activities relate to each other and how learning relates to marketing capability in firm performance has received limited attention.

In this study, we offer an expanded view of organizational learning in the context of early internationalizing firms, also known as born globals and international new ventures (INVs). Such firms provide an appropriate context as leading exemplars of successful international small and medium enterprises (SMEs) that challenge conventional internationalization theories by directly entering international markets at or near their founding (Oviatt and McDougall 1994; Knight and Cavusgil 1996). As young entities, they endure limited resources and, despite this, target

relatively uncertain and risky foreign markets (e.g., Freeman et al. 2006; Knight and Cavusgil 2004). Despite growth over two decades, however, research on early internationalization remains fragmented (e.g., Rialp et al. 2005; Weerawardena et al. 2007). In the extant literature on INVs, scholars have examined the role of market learning, but other learning types have not been explicitly addressed (Autio et al. 2000; McDougall et al. 2003).

We draw on the dynamic capabilities view (e.g., Teece 2007, 2012) to capture the set of capabilities that drive innovation and early internationalization. Dynamic capabilities are an appropriate theoretical foundation because of their role in developing new knowledge resources that enable firms to pursue value creation strategies (Eisenhardt and Martin 2000; Cepeda and Vera 2007; Zahra et al. 2006). Dynamic capabilities enhance the firm's ability to seize new opportunities and undertake challenging market entry (Teece 2007). However, dynamic capabilities research has emphasized established firms, overlooking new ventures and SMEs. Such firms typically employ dynamic capabilities to get established, achieve legitimacy, grow, survive, and reap the benefits of innovation (Sapienza et al. 2006).

We adapt the organizational subsystems view (Damanpour et al. 1989) to examine the interplay of dynamic capabilities and their role in early internationalization. We theorize that market-focused learning capability and marketing capability constitute a "market subsystem", and knowledge acquisition from network learning capability and internally focused learning capability constitute a "socio-technical subsystem." Exposition of these subsystems advances understanding of various types of organizational learning and how they relate to each other and to marketing capability, and in turn to early internationalization.

The dynamic capabilities view assigns a prominent role to key decision makers of the firm (Teece et al. 1997; Teece 2012) and facilitates examining founders' role in building and exploiting prerequisite knowledge resources for early internationalization. We argue that specific

dynamic capabilities are built and nurtured by founders who possess an international vision. These founders likely possess specific knowledge and experience, often arising from an international background (Oviatt and McDougall 2005), which drives them to value learning and to develop specific capabilities, to make internationalization succeed.

We contribute to the literature on organizational learning, early internationalization, and dynamic capabilities in the following ways. We identify that market learning is a necessary but not sufficient aspect of organizational learning. We explain how the dynamic capabilities view facilitates an enriched understanding of the development of new knowledge resources for enhanced organizational performance. In an INV context, we identify a set of capabilities based on market learning, internal learning, network learning, and marketing, which provides a comprehensive explanation of early internationalization. We argue these dynamic capabilities are driven by managers' international vision and operate in a dual organizational subsystem: the market subsystem and the socio-technical subsystem. This dual subsystem of dynamic capabilities drives innovation and thence early internationalization. We also highlight the need for future research in strategic marketing and internationalization in these topic areas. Our research responds to calls for marketing scholars to advance theory in international competitiveness (e.g., Hult 2012).

This study proceeds as follows. Initially, we present our conceptual foundation and how it relates to the early internationalizing firm context. We then discuss our methodology: a qualitative phase inspired our conceptual framework and hypothesized relationships, and a quantitative phase involving surveys in two countries helped confirm qualitative results. We conclude with discussion that holds implications for theory and practice, and directions for future research.

Conceptual foundation

We propose that early internationalization is enabled by the presence of a set of dynamic capabilities that are established from knowledge routines conveyed by founders who bring access to resources and managerial acumen to configure and deploy these resources appropriately in implementing strategy. As viewed by Mahoney (1995, p. 94) firm sustainable advantages are history dependent, and “managers’ past decisions, decision rules, and tacit understanding from experience are the basic genetics that firms possess.” We draw on the dynamic capabilities view of competitive strategy to capture the assortment of strategic capabilities that are built and nurtured by founding managers with an international vision that drives innovation and early internationalization.

We build on core theoretical propositions that have gained wider acceptance within the dynamic capability literature. First, there is a growing consensus within the literature that the role of dynamic capabilities in strategy formulation is to transform the firm’s knowledge resources to produce new configurations that can be applied to operationalize strategy (Cepeda and Vera 2007; Zahra et al. 2006). This view departs from early literature (e.g., Teece et al. 1997) that suggested a direct link between dynamic capabilities and firm competitive advantage that has not been empirically substantiated. We conceive dynamic capabilities to be the organizational and strategic routines by which firms *build, integrate and reconfigure* internal and external resources to address market opportunities (Helfat et al. 2007; Teece 2012; Zollo and Winter 2002). We adopt this definition to capture a firm’s capacity for developing new knowledge resource combinations to pursue various innovations that lay the foundation for early internationalization. Deliberate or intentional organizational learning routines are embedded in dynamic capabilities (Zollo and Winter 2002). Increased learning investments enable the firm to adapt its operations (Romme et al. 2010). Dynamic capabilities “determine the speed at, and degree to which, the

firm's particular resources can be aligned and realigned to match the requirements and opportunities of the business environment so as to generate sustained abnormal (positive) returns" (Teece 2012, p. 1395).

Three streams in the organizational learning literature inform our conceptualization and proposed interplay of learning capabilities. First, we employ literature on acquiring and strategically using knowledge from both within and outside the enterprise (e.g., Powell et al. 1996), and conjecture that these knowledge types exert a differential and synergistic impact on innovation. Whilst market learning (externally focused) captures knowledge about customers and competitors, network learning (externally focused) and internally focused learning capture diverse knowledge types essentially needed to deliver products and services that markets desire. Second, we leverage the literature that advocates adaptive firms must actively pursue both *exploration* (exploring knowledge, routines, and rules that might come to be known) and *exploitation* (using existing knowledge) in their learning (March 1991). Third, we employ literature on absorptive capacity (Cohen and Levinthal 1990) that stresses the firm's ability to value, assimilate, and apply external knowledge. Broadly, we argue that firms must maintain an appropriate balance in different types of knowledge acquisition, as an excessive investment in one type will lead to wasted resources and knowledge deficiencies within the firm. In addition, we conceive that marketing capability will support early internationalization by creating a capacity to build, integrate, and reconfigure strategic marketing tools and expertise to effectively identify, access, and deliver value in international markets (Weerawardena et al. 2007). As they operate in diverse (international) markets, managers acquire knowledge on marketing strategies and tactics (Zhou et al. 2012). In this way, learning serves to enhance vital marketing skills.

Our context is young, resource-poor early internationalizing firms for whom dynamic capabilities may be especially important because such businesses target diverse, risky foreign

markets (Luo 2000). Risks, costs, and uncertainties confront the nascent firm externally as a dearth of legitimacy with stakeholders and in target markets (Mudambi and Zahra 2007; Zhou et al. 2010). Such risks may be less constraining in new firms when founders develop appropriate capabilities (Autio et al. 2000) that enable building new, beneficial routines for operational effectiveness (Zhou et al. 2010). Appropriate capabilities help attenuate the liabilities of foreignness and newness in foreign markets (Oviatt and McDougall 1994).

Study 1: Qualitative research study

The literature on organizational learning combined with the dynamic capabilities view and the extant early internationalizing firm literature led us to an *a priori* identification of a set of capabilities that are present within early internationalizing firms. To determine how these capabilities relate to each other and in turn to innovation and early internationalization, we adopted a two-stage, multi-method design: first a qualitative study followed by a quantitative study (Creswell 2003). This design enabled us to develop the theoretical relationships to be identified for subsequent testing.

Data collection and analysis

The primary objective of the qualitative research study is to understand the dynamic and evolutionary processes of the firm (Kamakura et al. 2012), specifically to identify the organizational routines brought into the young firm by its founders, and how they exploited these routines to drive early internationalization. We conducted in-depth interviews with founding managers and CEOs of 14 early internationalizing firms in Australia (Table 1). We chose founders and senior managers with substantial knowledge of the organizational characteristics, strategy, and performance (Snow and Herbsminak 1980). Initial interviews lasted about 2.5 hours,

with multiple follow-up interviews over a two-year period. The firms are based in Australia, where McKinsey (McKinsey and AMC 1993) conducted some of the earliest studies on born global firms. The firms were drawn from both high and low-tech industries, with a mix of B2B and B2C firms. We also included start-ups that had not internationalized in the required time (within three years of founding), consistent with numerous studies (e.g., Madsen and Servais 1997), in order to maximize theoretical variation (Eisenhardt 1989).

Insert Table 1 about here

Consistent with Rindova et al. (2011) and Montealegre (2002), we conducted our analysis in several steps to iterate between data and possible theoretical conceptualization (Glaser and Strauss 1967). First, we developed an overall case narrative and chronological description (Miles and Huberman 1994) depicting the sequence in which organizational routines were built and nurtured by the founders. In this way, we traced the nascent firms' unlocking and aggregation of routines to pursue international market opportunities. The process of analysis was adapted from Glaser and Strauss (1967), Creswell (2003) and Yin (1994). In analyzing the data, we adopted several techniques to maintain appropriate focus on the research problem. We developed coding frames within and across cases and employed a formal matrix as an analytical tool to ensure pattern matching and effective categorization of data (Miles and Huberman 1994). Adopting the approaches of Rindova et al. (2011), we then developed a data structure analysis leading to specific theoretical insights. In an iterative process, we then related extant literature to the theoretical observations. Tying emergent theory to the extant literature enhances internal validity, generalizability and theoretical constructions (Eisenhardt 1989). We sought both conflicting and similar frameworks to build the new conceptual framework to be assessed in the quantitative study.

Findings

The following findings from this phase have important implications for the subsequent study.

First, the founders of the sampled early internationalizing firms were characterized by an international vision and an action orientation to build prerequisites for early internationalization.

The founders' prior employment displayed repetitive patterns of knowledge accumulation in market knowledge development, relationship-based knowledge development, technology and new product development, and international marketing. Second, we identified the unlocking and subsequent aggregation of prior routines with new combinations of knowledge through collective sense making (Pavlou and El Sawy 2011) aimed at exploiting a potential market opportunity. This paved the way for the emergence of dynamic capabilities, specifically, market focused learning, network learning, internally focused learning and marketing. This process revealed activities for the transformation of initial knowledge resources to new knowledge combinations, i.e., the build, integrate and reconfigure transformations. Interestingly, these transformation activities were not undertaken in a sequential manner as implied in the literature (Teece et al. 1997). Third, we identified how the firms harvested new knowledge configurations to develop technology- or knowledge-intensive product innovations, along with marketing and process innovations. Market-focused learning was observed but was identified as operating with marketing capability within a dual system including other organizational learning capabilities. Evidence suggested marketing capability tends to differentiate firms that internationalize early (within three years) from those that internationalize much later.

Subsystem framework of capabilities and innovation within an internationalization context

Our findings led us to develop a new conceptual framework with a set of sub-systems leading to innovation and, in turn, to early internationalization. Assigning a central role to innovation (e.g., Han et al. 1998; Hurley and Hult 1998), which we theorize as a key mediator, we build on two key propositions in the innovation literature. First, we adopt the *technical* and *administrative* (*non-technical*) typology of organizational innovation (Damanpour et al. 1989). This typology highlights an expanded role for innovation, beyond the traditional focus in the marketing literature which has tended to examine innovation as new product development. However, scholars have appealed for a richer perspective on innovation's role (Han et al. 1998; Hurley and Hult 1998; Varadarajan and Jayachandran 1999). Our findings suggest that innovation likely plays a key role in transforming learning into action, in our case, early internationalization. Second, we adapt the organizational subsystems framework (Damanpour et al. 1989) to capture the interplay of dynamic capabilities identified in the qualitative study as prerequisites for early internationalization. These dynamic capabilities are internally focused learning, network learning, market-focused learning and marketing. As noted earlier, whilst learning capabilities capture the development of new knowledge for innovation, marketing capability represents the firm's capacity to enter an international market in a speedy manner with the innovative offerings it has developed.

We extend and adapt the Damanpour et al. (1989) subsystems typology to the early internationalization context in three ways. First, Damanpour et al. (1989) primarily used the subsystems approach to examine the interplay between types of innovation. However, because we emphasize antecedent capabilities that drive innovation and early internationalization, we examine the interplay of knowledge processes leading to innovation. Second, a subsystem represents rules,

roles, procedures, and structures related to communication and exchange among organizational members, and between the environment and organizational members (Damanpour et al. 1989). Drawing on the dynamic capabilities view, we suggest that knowledge-building routines also should be included (Eisenhardt and Martin 2000; Zollo and Winter 2002). Third, Damanpour et al. (1989) examined technical and administrative innovation in large public libraries in which internal administrative knowledge processes are important. However, because we investigate early internationalizing firms in which external market focus is needed, we suggest including the market subsystem to accommodate market learning and marketing. Our two adapted subsystems comprise market and socio-technical subsystems. The market subsystem is primarily market-focused, whilst the socio-technical subsystem is focused on building the internal knowledge base. Our approach is consistent with the subsystems view that the focus of knowledge building activities will depend on the nature of the organization (Damanpour et al. 1989).

Our conceptual framework is indicated in Figure 1. The framework theorizes an interplay of capabilities with innovation that influences early internationalization. The capabilities are built and nurtured by founders with an international vision. The paper proceeds by developing a set of hypotheses, all of which relate to the early internationalizing firm.

Insert Figure 1 about here

Hypotheses development

International vision

Early internationalizing firms are characterized by a strong international orientation from their inception (Knight and Cavusgil 1996; Oviatt and McDougall 1994). Founders bring a strong international vision to such firms, rooted in their prior experiences and knowledge. We define

international vision as a managerial orientation, linked to a proactive culture, that supports actively exploring new business opportunities abroad, developing business in international markets, accruing resources and skills aimed at achieving goals in such markets, and committing to strategies appropriate for success in the markets (Knight 2000; Knight and Cavusgil 2004; Knight and Kim 2009; Sullivan Mort and Weerawardeena 2006). In our interviews, all managers displayed an international vision to build a firm that takes their nascent products to the international marketplace. We argue that international vision gives rise to learning, which in turn supports the development of dynamic capabilities (Knight 2000; Oviatt and McDougall 1994). For example, as they pursue international opportunities, internationally visionary firms likely develop a distinctive outlook and accompanying competences (e.g., McDougall et al. 2003; Sullivan Mort and Weerawardenena 2006).

In new venture firms lacking substantial resources, a proactive and visionary posture is needed to deal with risky and complex environments (e.g., Mudambi and Zahra 2007). Visionary leadership tends to promote shared mental models that encourage and facilitate learning and the use of new technologies aimed at achieving company goals (Senge 1990). Visionary leadership supports the development of a learning- and innovation-oriented culture (Slater and Narver 1995; Vera and Crossan 2004), as well as the creation of key capabilities (Helfat et al. 2007; Ireland et al. 2009). A capacity to learn helps span the psychic (Johanson and Vahlne 1977) and institutional distance (Schwens et al. 2011) between the home country and international contexts. As indicated in the case studies, we expect international vision is particularly useful in building, integrating and reconfiguring needed knowledge resources in the firm. In short, the presence of a strong international vision tends to nurture a company environment that supports and aligns capabilities that emphasize organizational learning.

Socio-technical subsystem

Internally focused learning capability Based on the dynamic capabilities view (Teece 2012), we define an internally focused learning capability as the firm's capacity to build, integrate and reconfigure technical and social (non-technical) knowledge generated through internal sources. The sampled firms actively pursued learning from internal sources. This capability captures all the experimental learning of the firm (Weerawardena et al. 2007), including traditional R&D and technological learning (Zahra et al. 2000) and social (non-technological) learning, which engenders innovation and the firm's ability to respond to evolving conditions in its external environment (Nelson and Winter 1982).

Internally focused learning supports superior organizational performance in different ways. For one, firms that emphasize technological learning are better at adapting to and growing in new markets (Autio et al. 2000; Liesch and Knight 1999). Use of knowledge-based internally determined innovation strategies is associated with international growth of the firm (Autio et al. 2000), enabling many new firms to perform above their counterparts in international activities, often evidenced as early entry into foreign markets. The capacity for precocious expansion to international markets is associated with global technology competence (Rialp et al. 2005; Zahra et al. 2000) and unique product development (Knight and Cavusgil 2004), the latter being an outcome of their technology competence. Internal learning is one of the key sources of technological competence that leads to increased absorptive capacity (Cohen and Levinthal 1990), which is a critical element of the internal resource base of the early internationalizing firm (Rialp et al. 2005). Thus, we frame:

H1: The founders' international vision will be positively related to the internally focused learning capability of the firm.

Network learning capability We define network learning capability as the firm's capacity to build, integrate and reconfigure technical and non-technical knowledge generated through external links and institutions. Early internationalizing firms, particularly small new firms, are vulnerable because they usually possess few financial and tangible resources. Many such firms depend on a single product that they commercialize in a lead market first, and often seek partners who complement their own resources for operational effectiveness (Oviatt and McDougall 1994; Selnes and Sallis 2003). Networks facilitate knowledge acquisition relevant to internationalization and can reduce some uncertainties of operating in foreign markets (Freeman et al. 2006; Sullivan Mort and Weerawardena, 2006). The sampled firms actively pursued learning from network sources. After discovering and perceiving the technologies that enable internationalization, entrepreneurs establish international networks to explore where and how quickly the opportunity can be exploited (Oviatt and McDougall 1994, 2005). As noted earlier, knowledge intensity associated with vision is a prerequisite for aggressive opportunity seeking (Oviatt and McDougall 2005), and network learning is one of the externally focused learning capabilities that contribute to the knowledge intensity of the new firm in its early internationalizing activities. A network learning capability captures the firm's abilities, through the vision and precocious experiences of the founders, to access existing networks and to develop new ones for internationalization. We are led to hypothesize:

H2: The founders' international vision will be positively related to a network learning capability of the firm.

Though most firms, particularly large ones, internally develop much of the knowledge used in innovation, small start-up firms do not possess all the inputs required for successful and

continuous technology development. This compels them to seek external technological knowledge (Rosenkopf and Nerkar 2001). Because exploiting knowledge generated by others is a function of each organization's access to this knowledge, idiosyncratic differences in knowledge retrieval arise across firms (Almeida et al. 2003). Internal and external learning activities are related and often complementary (Powell et al. 1996) since the ability to integrate and make effective use of externally acquired knowledge depends partly on the amount and quality of the firm's prior knowledge (Cohen and Levinthal 1990). However, the firm's awareness of where useful complementary expertise resides within and outside the organization is an important prerequisite for this purpose. As Cohen and Levinthal (1990, p.133) identify, "this is the sort of knowledge of who knows what, who can help with what problem, or who can exploit new information." Thus, we frame:

H3: The firm's network learning capability will be positively related to an internally focused learning capability of the firm.

Market subsystem

Market-focused learning capability We define market-focused learning capability as the firm's capacity to build, integrate and reconfigure technical and non-technical knowledge acquired from the firm's customers and competitors. An ability to integrate market information into actionable knowledge that management can apply into international markets (Knight and Liesch 2002; Madsen and Servais 1997) is an important capability within the firm. The acquisition of experiential knowledge and its application to a target market long has been seen as a prerequisite for successful internationalization (Johanson and Vahlne 1977). The sampled firms displayed the strategic pursuit of acquiring knowledge from markets. Thus, identifying niches that match the firm's products and services to customer expectations facilitates early internationalization (Crick

2009). As found in our initial qualitative phase, in their product development stage, high-tech early internationalizers often work with lead customers to ensure their products are compatible with customer needs (Madsen and Servais 1997). Because entrepreneurship can support not only outdoing competitors in new product development but also anticipating customer needs, market-focused learning is an important characteristic of entrepreneurial firms (Weerawardena 2003a). We conjecture that founders' international vision, as a manifestation of prior international exposure (Madsen and Servais 1997; McDougall et al. 2003) and network knowledge (Oviatt and McDougall 2005), facilitates the presence of greater market knowledge within the early internationalizing firm. We hypothesize:

H4: The founders' international vision will be positively related to the firm's market-focused learning capability.

Marketing capability Marketing capability (Day 1994b; Knight 2000; Weerawardena 2003b) is the result of an integrative process designed to apply the collective knowledge, skills and resources of the firm to the market-related needs of the enterprise. We define marketing capability as the firm's capacity to build, integrate and reconfigure strategic marketing tools and acumen for effectively identifying and accessing international markets and delivering value to these markets. As construed in this study, marketing capability refers to developing products or services with global appeal that create value for targeted buyers; skillfully marketing communications to promote the firm's offerings; integrating marketing resources and skills to best meet customer needs; and adjusting marketing resources and skills to address evolving customer and competitor circumstances (Dutta et al. 1999; Weerawardena 2003b). Prior experiences of the firm's founders provide a foundation for organizational capabilities that enables the firm to position and fit products for international markets, to communicate the value of products, and to price and

distribute products skillfully (Madsen and Servais 1997; Weerawardena 2003b). Our case study findings indicate that early internationalizing firms build and nurture distinctive marketing capabilities from the vision and experiences of the founders that enable them to effectively access and penetrate international markets early with innovative products. Thus, we frame:

H5: The founders' international vision will be positively related to the firm's marketing capability.

Organizational learning influences organizational behavior by action-oriented use that is the direct application of knowledge to solve a problem, and knowledge-enhancing use that influences managerial perspectives on problems (Menon and Varadarajan 1992; Slater and Narver 1995). We argue that market-focused learning enhances the marketing capability of the organization. As widely accepted approaches to target marketing suggest (Day 1994b; Slater and Narver 1995), understanding market characteristics is a prerequisite for effective use of marketing mix strategies to reach the desired market segment. Prior international experience enables early internationalizing firms to effectively reach intended markets (Knight 2000). Market-focused learning enriches marketing capability, better enabling the firm to reach niche markets. Our approach is consistent with Hult et al. (2005, p.1174), who argue that market learning (market information processing) does not directly influence performance but "allows the firm to enact better actions, which in turn should enhance performance." We hypothesize:

H6: The firm's market-focused learning capability will be positively related to the firm's marketing capability.

The role of network relationships in building a marketing capability in the firm has received substantial attention in the literature (Srivastava et al. 2001). Relationship-based assets

such as strategic partnerships, networks and eco-system relationships can be leveraged to deliver greater customer value through market-based processes such as new product development (Srivastava et al. 2001). Networking capability provides access to potential customers and members of the international value chain (Weerawardena et al. 2007; Sullivan Mort and Weerawardena 2006). We frame:

H7: The firm's network learning capability will be positively related to the firm's marketing capability.

Innovation performance

Drawing on Damanpour et al. (1989) and Weerawardena (2003 a, b), we define innovation performance as adoption of new and value-adding changes to the products (goods and services), processes, and market establishment activities within a firm. As conceptualized in our paper, dynamic capabilities produce new knowledge resources facilitating innovation in its various forms, such as technological, marketing, organizational and managerial innovations (Han et al. 1998; Kim and Atuahene-Gima 2010). Innovation can be viewed as a knowledge transformation process (Mahoney 1995). Strategic efforts to acquire and respond to market-focused learning enhance the firm's ability to create and deliver superior customer value by developing various product-market innovations (Slater and Narver 1995). Because knowledge of market preferences increases the degree of compatibility of new products with customer needs, it is likely to enhance the adoption and success of innovations (Kim and Atuahene-Gima 2010). Organizational learning that results from inward- and outward- focused capabilities gives rise to innovations that drive company performance (Zheng et al. 2005).

Innovation is vital because it gives rise to new products and services, and new ways of doing business, which provide the nascent firm with the means to undertake early

internationalization. Early internationalization occurs in a wide variety of product and service firms, in both the high- and low-tech sectors (Rialp et al. 2005). Our case study findings indicate that innovation in an early internationalizing firm context is not restricted to technologically advanced products but is accompanied by “soft” innovation in marketing systems and process innovations internal to the firm. Consistent with organizational learning approaches to innovation (Cohen and Levinthal 1990; Hartman et al. 1994), we argue that externally and internally focused learning capabilities support the ability of new venture firms to develop products and markets, and novel ways of doing business, which are requisite to successful early internationalization. Marketing capability allows such firms to transform market-generated knowledge into resources, skills and activities for meeting customer needs. Product, process and marketing innovations provide the new firm with a leading edge in its internationalization process activities. Thus, we offer:

H8: There will be a positive relationship between a firm’s internally focused learning capability and its innovation performance.

H9: There will be a positive relationship between a firm’s network learning capability and its innovation performance.

H10: There will be a positive relationship between a firm’s marketing capability and its innovation performance.

H11: There will be a positive relationship between a firm’s market-focused learning capability and its innovation performance.

Outcome construct: Early internationalization

Building on Pla-Barber and Escribá-Esteve (2006), *early internationalization* is defined here as the display of the following firm behaviors in international market entry: earliness (within a short

period of time from firm foundation), scope (entry to multiple markets) and extent (increased revenue from markets entered). Superior firm performance hinges on the ability of management to harvest new knowledge to develop a market offering that matches with market opportunity. In our context, innovation is associated with the development or improvement of products and services, and new ways of doing business in terms of the firm's production processes, and market establishment activities (Han et al. 1998; Weerawardena 2003b). For early internationalizing firms, it implies entering new international markets with new or established goods and services. Product uniqueness or knowledge-intensive products (Bloodgood et al. 1996; Weerawardena et al. 2007) is a prerequisite for early internationalization. Innovation based on organizational knowledge intensity supports the firm's ability to adapt efficiently to foreign environments and to perceive opportunities for foreign expansion (Autio et al. 2000; Bloodgood et al. 1996). An innovation-oriented posture may be necessary in firms that, in the face of relatively limited resources, take the initiative to pursue new opportunities in complex markets, characterized by uncertainty and risk (Knight and Cavusgil 2004). The concurrent adoption of both technical and non-technical innovation tends to enhance organizational performance (Damanpour et al. 1989; Han et al. 1998). Similarly, we argue that broad-based innovation supports early internationalization.

Likewise, we surmise that marketing capability is associated with early internationalization. Marketing capability engenders superior performance in domestic (e.g., Varadarajan and Jayachandran 1999) and internationalizing firms (Zhou et al. 2012). As our case findings suggest, marketing capability distinguishes early internationalizers from others in the sample who took an extra few years in spite of the presence of founders' international vision and product advantage within such firms. Marketing prowess provides the foundation through which the firm interacts with diverse foreign markets. Firms with a strong marketing capability create

specific marketing strategies aimed at overcoming the challenges inherent in foreign markets (Knight and Cavusgil 2004). Rialp et al. (2005) confirm that early internationalization is associated with the creation and marketing of products of high value. Therefore, our final hypotheses are:

H12: Innovation performance will be positively related to the firm's early internationalization.

H13: Marketing capability will be positively related to the firm's early internationalization.

Study 2: Quantitative research study

Questionnaire and sampling frames

Having built the conceptual framework based on theoretical observations from the qualitative study and the enfolded literature, we tested the hypotheses in a quantitative study. After framing a questionnaire, we pilot tested it to improve its clarity and meaningfulness to managers. We then collected data in a mail survey from firms in Australia and the United States that had internationalized early. CEOs/senior managers were used as the key informants. Respondents were directed to recall and reflect on the "learning and the new ways of doing things that helped make the firm what it is today." As an additional measure (Kumar et al. 1993), a self-assessment of knowledge of firm activities was adopted, with the mean response on a five-point Likert scale ($\mu = 4.78$; $\sigma = 1.43$, Australia, and $\mu = 4.77$; $\sigma = 0.614$, U.S.), giving evidence of knowledge of the firm, its vision, and its growth strategies.

Mail survey

Our survey in Australia of 11,000 exporting firms yielded 1,046 completed responses, a response rate of 9.5%. From these firms, we identified a subset that began exporting within three years of

their founding and generated 25% or more of total earnings from exports, qualifiers from the born global literature (Madsen and Servais 1997) that we adopted as widely accepted, robust evidence of early internationalization. This step resulted in useable responses from 201 firms. On average, these firms employed just over 100 employees, and had annual sales of about AUD71 million. We used the same method and criteria to conduct a survey in the United States, yielding a usable sample of 234 firms, a response rate of 13.4%. On average, the U.S. firms employed about 190 employees, and had annual sales of about AUD40 million. In both samples, non-response bias was assessed by comparing early- and late-responding firms (Armstrong and Overton 1977). We then conducted Mann-Whitney U-tests between the two groups, using demographic and non-demographic variables. Findings revealed no significant differences, which confirmed non-response bias was not a factor affecting study results.

Measurement issues

We undertook several steps to minimize common method variance (CMV). First, we used multiple items to measure each construct, helping dissipate short-term memory effects related to a previous scale(s) (Harrison et al. 1996). Second, we emphasized distinctive constructs in terms of face validity (Harrison et al. 1996). Third, we used a multiple item measure to separate the learning capability variables from the dependent variables in the questionnaire, as a means of reducing the tendency of respondents to rely on prior responses to answer questions (Podsakoff et al. 2003). Fourth, in post-hoc analyses, we assessed possible CMV using Harman's single-factor test (Harman 1976). The test generated a nine-factor solution (with eigenvalues >1.0) that accounts for approximately 66% of the total variance in the Australian sample, with the first factor accounting for only 30% of the total variance. In the U.S. sample, the Harman's test generated an eight-factor solution (with eigenvalues >1.0) that accounted for approximately 65% variance, of

which the first factor explained 29% of the total variance. Because a single factor solution did not emerge and the first factor was not associated with the majority of variance, systematic variance in the study measures was not found to be an issue (Podsakoff and Organ 1986). In a fifth step, we used the marker-variable technique suggested by Malhotra et al. (2006). Our testing suggests that none of the original correlations of relevance appeared to differ significantly from the CMV-adjusted estimates (c.f. Bollen et al. 1989), which again suggested common method bias was not a problem affecting results of the study.

We also took steps to ensure the equivalence of the samples between Australia and the United States. Prior to collecting the U.S. data, the instrument was tested for face validity and conceptual/calibration equivalence on a small group of U.S. respondents (Mullen 1995). A comparison of the construct reliability scores (see Table 3 and Table 4) showed consistency, thereby suggesting the measures are fairly equal across the two samples (c.f. Singh 1995). We employed the Durbin-Wu-Hausman test to check for endogeneity in our outcome variable, early internationalisation. Our testing showed consistent regression results in both samples, suggesting that endogeneity bias may not be an issue in the current study.

Measures

To assess constructs, we employed a combination of new scales and scales from the literature modified appropriately to suit our context. Except as noted, all measures were five-point Likert scales anchored by “strongly disagree” and “strongly agree.” All construct measures used in our model are unidimensional and reflective (Jarvis et al. 2003). Our six-item scale for *international vision* was adapted from Knight and Kim’s (2009) measure of international orientation. The four *dynamic capability* constructs within the two subsystems were operationalized with items representing each of the activities: build, integrate and reconfigure (Teece 2012). The *innovation*

construct was operationalized with three sub-dimensions, viz., *product or service innovations*, *process innovations*, and *market establishment innovations*, using two items to assess each sub-dimension. The measure was adapted from Weerawardena (2003a, b) and is consistent with prior research (e.g., Damanpour et al. 1989). Respondents were asked to assess new ideas adopted by their firm, as well as the scope of innovations (anchored by “few” to “many”), and the degree of innovations (anchored by “incremental” to “radical”). The outcome construct, *early internationalization*, was measured using seven items, based on Pla-Barber and Escribá-Esteve (2006). This scale measures three dimensions, *earliness*, *scope* and *extent* of internationalization. The battery of items is presented in Table 2.

Insert Table 2 about here

Results

We analyzed our data using partial least squares in structural equations modeling (PLS-SEM) (Hair et al. 2010), which simultaneously estimates measurement models and the structural model. Consistent with prior research, the use of PLS-SEM is appropriate given the small sample sizes relative to the number of parameters to be estimated [recommended parameters to cases ratio is 1:5 (Kline 2005)] and a non-normal distribution of data in both samples with model complexity (as PLS is not constrained by model identification problems) (Henseler et al. 2009). In addition, given the infancy of quantitative research within the dynamic capabilities area and the use of newly developed measures in the model, the study contributes to theory development (Hair et al. 2010), for which the simultaneous estimation undertaken in PLS is justified.

Exploratory factor analysis with oblimin rotation was undertaken for all the measures. Summary statistics are shown in Tables 3 and 4 for the Australian and U.S. samples, respectively. As shown, the analyses support unidimensionality for the latent constructs in the model. The

Average Variance Extracted (AVE) for all measures was 50% or more, demonstrating internal stability and satisfactory convergent validity (Fornell and Larcker 1981). The factor loadings for all constructs were acceptable (> 0.6). Satisfactory Cronbach's alpha and composite reliability estimates (> 0.7) indicated high levels of construct reliability. Construct discriminant validity was assessed by comparing the variance extracted estimate for any two constructs with the square of the correlation estimate between these two constructs. As shown in Tables 3 and 4, the squared correlation estimates were lower than the variance extracted estimates for each construct, establishing discriminant validity (Fornell and Larcker 1981).

Model fit The model fit was assessed by examining the model fit indices provided by PLS, namely, the average path coefficient (APC), average R-squared (ARS), and average variance inflation factor (AVIF). For the APC and ARS, it is recommended that the p -values are lower than 0.05; AVIF is recommended to be lower than 5 (Kock 2012). The fit indices for the two samples: Australian sample [APC = 0.291 ($p < 0.001$); ARS = 0.325 ($p < 0.001$); AVIF = 1.581]; U.S. sample [APC = 0.334 ($p < 0.001$); ARS = 0.383 ($p < 0.001$); AVIF = 1.609] indicate a good fit of the models to the data. VIF values for both samples (< 2) suggest multicollinearity was not an issue (Hair et al. 2009). We also computed the Goodness of Fit (GOF) index discussed by Tenenhaus et al. (2005), a global fit index that simultaneously assesses the fit of the outer measurement and inner-structural models in PLS. The thresholds for GOF are 0.1, 0.25, and 0.36 for small, medium, and large effect sizes (Wetzels et al. 2009). The calculated GOF [Australian sample = 0.43; U.S. sample = 0.45] suggests robust fit of the model to the data. Finally, the predictive relevance of the model was assessed by examining the Stone-Geisser Q^2 coefficient (c.f. O'Cass and Weerawardena 2010). The Q^2 coefficient is a nonparametric measure and represents how well the observed values are reconstructed by the model and the model

parameters. The calculated Q^2 coefficients for all endogenous variables included in both the models were greater than zero, suggesting acceptable predictive validity for both models.

The model explained 28% of the variance in the outcome construct in the Australian sample and 39% of the variance in the U.S. sample. Finally, we controlled for firm age, size, and export intensity, comparing the base model to the main effects model. For the Australian sample, the base model (control variables only) explained about 2.5% of the variance in the early internationalization (EI) construct. The main effects model along with the control variables explained 29.3% variance in the EI construct. For the U.S. sample, the base model explained about 4.3% of the variance in the EI construct. The main effects model along with the control variables explained 39.3% variance in the EI construct. Except for export intensity in the Australian sample, none of the control variables are statistically significant when included in the main effects model.

Insert Tables 3 and 4 about here

The Australian and U.S. samples compared Figures 2 and 3 reveal the results of hypotheses testing for the Australian and U.S. samples, respectively. The figures show the path coefficients between the latent constructs and the results of the structural models. The path coefficients for the constructs are expressed in standardized form with the associated t-values in the parentheses.

Insert Figures 2 and 3 about here

For both samples, results revealed significant positive relationships between *international vision* and the dynamic capabilities, viz., *internally focused learning capability* [H1: $\beta_{\text{Aus}} = 0.37$; $t = 5.44$ ($p < .001$); $\beta_{\text{US}} = 0.12$; $t = 1.71$ ($p < .05$)]; *network learning capability* [H2: $\beta_{\text{Aus}} = 0.31$; $t = 4.73$ ($p < .001$); $\beta_{\text{US}} = 0.28$; $t = 2.97$ ($p < .01$)]; *market-focused learning capability* [H4: $\beta_{\text{Aus}} = 0.54$; $t = 10.0$ ($p < .001$); $\beta_{\text{US}} = 0.65$; $t = 10.06$ ($p < .001$)]; and *marketing capability* [H5: $\beta_{\text{Aus}} =$

0.30; $t = 4.70$ ($p < .001$); $\beta_{US} = 0.35$; $t = 4.03$ ($p < .001$)). Similarly, both samples supported the hypothesized inter-play between capabilities, viz., *market-focused learning capability* $\rightarrow$ *marketing capability* [H6: $\beta_{Aus} = 0.39$; $t = 6.20$ ($p < .001$); $\beta_{US} = 0.48$; $t = 4.16$ ($p < .001$)]; *network learning capability* $\rightarrow$ *internally focused learning capability* [H3: $\beta_{Aus} = 0.31$; $t = 5.49$ ($p < .001$); $\beta_{US} = 0.67$; $t = 12.63$ ($p < .001$)] except for *network learning capability* $\rightarrow$ *marketing capability* in the U.S. sample [H7: $\beta_{Aus} = 0.23$; $t = 3.88$ ($p < .001$); $\beta_{US} = 0.009$; $t = 0.141$ ($p > .05$)]. With the exception of *network learning capability* $\rightarrow$ *innovation* in the Australian sample [H9: $\beta_{Aus} = 0.07$; $t = 0.964$ ($p > 0.05$)], dynamic capabilities were found to have a significant positive relationship with *innovation* in both samples, i.e., *internally focused learning capability* $\rightarrow$ *innovation* [H8: $\beta_{Aus} = 0.42$; $t = 5.95$ ($p < .001$); $\beta_{US} = 0.18$; $t = 2.49$ ($p < .01$)]; *network learning capability* $\rightarrow$ *innovation* [H9: $\beta_{US} = 0.28$; $t = 3.31$ ($p < .001$)]; *marketing capability* $\rightarrow$ *innovation* [H10: $\beta_{Aus} = 0.15$; $t = 1.75$ ($p < .05$); $\beta_{US} = 0.19$; $t = 2.47$ ($p < .01$)]. The hypothesized relationship between *market-focused learning capability* $\rightarrow$ *innovation* was supported only in the U.S. sample [H11: $\beta_{Aus} = 0.13$; $t = 1.51$ ($p > .05$); $\beta_{US} = 0.39$; $t = 4.01$ ($p < .001$)]. Finally, *innovation* and *marketing capability* respectively had significant positive relationships with the outcome construct, *early internationalization*, that is, [H12: $\beta_{Aus} = 0.20$; $t = 2.53$ ($p < .05$); $\beta_{US} = 0.36$; $t = 4.90$ ($p < .001$)] and [H13: $\beta_{Aus} = 0.40$; $t = 5.04$ ($p < .001$); $\beta_{US} = 0.36$; $t = 5.04$ ($p < .001$)] respectively.

Post hoc exploration We conducted post hoc analyses to test for additional curvilinear effects amongst the constructs in the model. Results showed that international vision² demonstrates a significant positive effect on network focused learning ($\beta = 0.785$; $p < .01$) and market-focused learning ($\beta = 0.391$; $p < .01$) in the Australian sample, but not the U.S. sample: network-focused learning ($\beta = -0.088$; ns) and market-focused learning ($\beta = -0.373$; ns) respectively. Similarly,

network-focused learning² demonstrates a significant positive effect on innovation ($\beta = 0.242$; $p < .05$) in the Australian sample, but not the U.S. sample ($\beta = 0.040$; ns) and marketing capability ($\beta = 0.366$; $p < .05$) in the U.S. sample, but not the Australian sample ($\beta = 0.055$; ns). Finally, somewhat surprisingly, the results suggest a significant negative effect of network-focused learning² on internally focused learning ($\beta = -0.623$; $p < .001$) in the U.S. sample, but not the Australian sample.

We also explored for interaction effects in our model. Results for the U.S. sample suggest that while the interaction between international vision and network learning may enhance internally focused learning ($\beta = 0.206$; $p < .05$), it may diminish market-focused learning ($\beta = -0.466$; $p < .05$). Similarly, the interaction of international vision and market learning demonstrates a significant negative effect on network learning ($\beta = -0.444$; $p < .05$) for the U.S. sample, but not the Australian sample. Correspondingly, for the Australian sample, the interaction of international vision and internal learning may enhance its marketing capability ($\beta = 0.121$; $p < .05$).

Overall, in addition to our general findings, the post hoc findings provide enhanced insights beyond the linear relationships. The curvilinear effects indicate that high levels of international vision may have positive effects on network-focused learning and market-focused learning, and high levels of network-focused learning may have positive effects on innovation for Australian firms. This suggests the need to nurture sophisticated levels of international vision to optimize these relationships. Similarly, in the U.S. context, managers should be alerted to take advantage of the synergistic effects of higher levels of network-focused learning on marketing capability, while avoiding the antagonistic effects on internally focused learning capability, due to absorptive capacity constraints (Cohen and Levinthal 1990).

A somewhat different and complex picture emerges with our interaction effects testing. Australian firms that have superior internal learning resources may benefit from an enhanced link

between international vision and marketing capability. For U.S. firms the situation is surprisingly more complex. Network learning moderates the relationship between international vision and market-focused learning, and vice versa in U.S. firms. This indicates that pursuing one form of external learning may have a detrimental effect on the other, perhaps signaling a more balanced approach to be adopted by managers. Overall, although the non-linear relationships are not the focus of our current study, collectively, the analyses provide research insights on differences between the two samples, an area worthy of further investigation.

Discussion

We have argued that the strategic marketing and internationalization literatures provide only a partial explanation of the conjectured relationship between organizational learning and firm performance. We were motivated by the fact that both the strategic marketing and internationalization literatures have primarily reported on the role of market learning, paying lesser attention to other types of learning and little attention to the relationship between learning and the firm's marketing capability. Adopting a multi-method, two-stage research design, we develop a more comprehensive explanation of the role of organizational learning using the early internationalizing firm context. Building on extant INV literature and drawing on the organizational learning and dynamic capability literature we determined *a priori* identification of the presence of a set of dynamic capabilities that facilitates early internationalization. The extensive casework that followed suggested a complex interplay of these capabilities, which we built into an organizational subsystems framework: the market subsystem that includes market learning capability and marketing capability, and a socio-technical subsystem of internal and network learning capabilities. This framework facilitates a deeper examination of the

contribution of the dynamic capability structure within a firm to innovation and early internationalization. The founders' international vision was found to be a key driver for building the prerequisites for early internationalization. Assembling these constructs and their observed relationships, we offered a conceptual framework which we tested in two types of markets, a small domestic market (Australia) and a large domestic market (U.S.).

The results in general support our hypothesized relationships. Market-focused learning and marketing capability operate in one subsystem that not only allows learning from markets but also relates this learning to enhance marketing capability, enabling firms to rapidly take their products to market. Similarly, the socio-technical subsystem, involving an internal learning capability and a network learning capability, provides new knowledge configurations to develop cutting edge market offerings to address customer needs identified through market learning. In a substantial departure from past approaches to learning-innovation-performance, the study's subsystems approach facilitates understanding the complex relationships between capabilities and their contribution to innovation and performance outcome: early internationalization. Our results are also robust to firm age, size and intensity of exporting.

The comparative structural models provide deeper insights into emphasis placed on deploying dynamic capabilities by the sampled firms in the two country markets. The structural model for Australia (Figure 2) confirmed all but two hypotheses. We did not find support for the hypothesized effect of network learning and market-focused learning capabilities on innovation (H9 and H11) in the Australian sample. However, we found that network learning capability is likely to take effect through the internally focused learning and marketing capabilities. Lack of support for H9 may indicate that network learning capability is deployed to augment the early internationalizing firm's internal learning processes as these firms are short on resources and do not have access to the full suite of internal business functions for operational effectiveness.

Shortfalls in these business functions may be met through leveraging on partner firms and other organizations. Similarly, lack of support for H11 in Australia implies that Australian firms primarily deploy market-based knowledge to enrich marketing capability, which in turn impacts innovation. Our structural model for the United States (Figure 3) confirmed all but one of the hypotheses (H7), the link from network learning to marketing capability. Lack of support for this hypothesis may indicate that U.S. firms primarily use market-focused learning to enrich marketing capabilities and not network learning as their Australian counterparts do.

A closer examination of the path values between the founders' vision and the dual subsystem of capabilities across the two samples provides further insights into this relationship, with direct paths from international vision to the capability set significant and positive in both samples (H1, H2, H4, H5). As indicated in the comparative structural models (Figures 2 and 3), the Australian firms seem to rely more heavily on internally focused knowledge development linking to innovation (H8) than do their U.S. counterparts [$(\beta_{\text{Aus}} = 0.42; t = 5.95 (p < .001); \beta_{\text{US}} = 0.18; t = 2.49 (p < .01))$]. The Australian firms rely less on network learning links to innovation (H9) than their U.S. counterparts [$(\beta_{\text{Aus}} = 0.07; t = 0.964 (\text{ns}); \beta_{\text{US}} = 0.28; t = 3.31 (p < .001))$]. It is possible that due to the geographic isolation of Australia, and wide dispersion of firms, opportunities for network learning in Australian firms are more restricted than in the U.S. where the large country market, denser agglomeration of firms, and greater integration with neighboring markets may provide more opportunities for network learning. In summary, we observe that when it comes to the set of links to innovation, Australian firms seem to rely more heavily on internally developed knowledge and marketing capability, whereas their U.S. counterparts use all three learning capabilities and marketing capability to drive innovation.

Regarding market-focused learning capability, we find a somewhat complex relationship to marketing capability and innovation across the two sets of firms. While the importance of a

market-focused learning capability to the dual system of capabilities is confirmed, its mode of effect differs across the two samples. In the Australian sample there is no direct relationship between market-focused learning and innovation; the effect of market-focused learning is captured through marketing capability's links to innovation. In contrast, in the U.S. firms, market-focused learning capability is directly related to innovation, as well as impacting marketing capability. Overall, U.S. firms appear to take greater advantage of all types of learning, whereas the Australian firms tend to be less focused on market-driven innovation. Findings from both samples revealed that, among new firms that internationalize early, international vision is instrumental in early internationalization. International vision drives the development of dynamic capabilities that, through their inter-relationships, create new knowledge configurations needed by firms to undertake multiple forms of innovation that engender early internationalization.

As intended, our findings offer an expanded explanation of organization learning's suggested link to firm performance. The findings also provide support to resolve the controversy concerning the role of market learning in determining firm performance. We find that market learning operates within a complex interplay of two subsystems of capabilities that leads to innovation and achieving organizational goals. The dynamic capabilities view that was adopted in our study facilitated examining (a) the role of the founders' international vision in building and nurturing a strategic set of dynamic capabilities and (b) the role of dynamic capabilities in developing new knowledge resource configurations needed for innovation and enhanced goal achievement.

Our conceptualization of the strategic organizational learning and marketing activities undertaken by the firm as dynamic capabilities contributes to both the organizational learning and dynamic capability literatures. Whilst the organization learning theorists suggest that organizational learning involves several activities/stages, i.e., *knowledge acquisition*,

dissemination, shared interpretation and organizational memory (Huber 1991; Slater and Narver 1995), by adopting the dynamic capabilities view, encompassing the *build, integrate* and *reconfigure* routines, we provide a better explanation of how to tie knowledge transformation activities to a strategic purpose within the firm. In spite of wider discussion within the literature (Helfat et al. 2007; Teece et al. 1997; Teece 2012), this conceptualization has escaped empirical scrutiny. While our research context was early internationalizing firms, we believe different firms will build different assortments of capabilities and harvest them differently. Interestingly, our case study findings suggest that the build, integrate and reconfigure activities may not occur in a sequential manner as implied by literature. Firms may again and again build new knowledge in their pursuit of appropriate product–market fit.

This research clarifies the important mediating role of innovation embedded in a system of relationships leading to early internationalization. In addition to new product development, we find that innovation promotes new processes and market establishment. Overall, our findings suggest that market learning is only one source of organizational learning that operates in a complex dual system of capabilities, along with innovation, providing prerequisites for realizing company goals. For the strategic marketing literature, the findings pave the way for future researchers to further investigate complexities involved in the interplay of market learning with other capabilities and contingency factors facilitating or constraining such relationships. For internationalization theory, our unified framework contributes towards the long felt need for a theoretical framework that provides a well-founded and comprehensive explanation of antecedents that drive early internationalization.

Implications for managers

For practicing managers, particularly of resource-constrained small firms aspiring to expand abroad, our findings provide a feasible path to early internationalization. These firms must build and nurture a set of dynamic capabilities that facilitates the acquisition of technological, market and marketing knowledge. Our findings suggest firms must actively engage in building networks with potential sources for technology acquisition, which is suggested as a way of accessing much needed technological knowledge for small firms. Similarly, internally focused learning is a prerequisite for building the firm's internal knowledge base. Access to external knowledge will guide the extent of the internally generated knowledge. In addition to learning about target markets, managers should recognize that building marketing capabilities is critical for pursuing innovation and for taking innovative product to international markets. As our qualitative findings suggest, market capability differentiates early internationalizing firms from those that take extra time to internationalize, despite possessing product advantages. Our conceptualization of dynamic capabilities in terms of the three activities – *build*, *integrate* and *reconfigure* – provides useful insights for practitioners in building and nurturing dynamic capabilities. These activities do not necessarily occur in sequence. Moreover, managers will need to continuously develop new knowledge to develop the technology content of their products, to match target customers' expectations. Managers also should recognize that multiple sources of innovation are important and can facilitate the firm's efforts to internationalize early.

Limitations and further research directions

Our research relied on accurate recall of knowledgeable insiders in firms surveyed at a particular moment in time. The two-stage research design with in-depth case studies enabled us to partially overcome this limitation, as well as to uncover key processes in firm foundation and

internationalization, and to elucidate relevant theoretical insights. We specifically selected case firms in which the founders were still involved, or are still accessible, to share their journey. The approach allowed us to develop a survey instrument with constructs using language descriptions that were relevant to the issues under investigation. Using a cross-sectional survey to clarify dynamic capability activities is consistent with past approaches to operationalize market learning (Hult et al. 2005; Jaworski and Kohli 1993). The measure of knowledge that we administered with the survey allowed increased confidence that we had contacted the appropriate survey respondents. For future research, we recommend a time series based design to investigate these issues further. Our model was tested using two country samples, one with a large domestic market (U.S.) and the other with a small domestic market (Australia). However, this captures only some of the natural variation in markets. Future research should be conducted in emerging and transitional markets to further validate the dynamics of the model.

Few firms possess all the capabilities required to achieve a competitive advantage. Arguably, developing various learning capabilities involves a tradeoff in terms of which capabilities to emphasize for development. Scholars might consider exploring how these decisions are made. As our post hoc exploration suggests, more needs to be understood about the optimization of knowledge configurations and how such efforts interact with international vision. Similarly, future research might explore the relative importance of these different learning strategies and the capability gaps that give rise to delayed internationalization. Further study may find that different types of learning are more important in some innovation contexts than in others and that firms must invest in particular kinds of learning in order to achieve a particular innovation outcome.

It is also important to consider there could be a reverse causal sequence or feedback loops within the model. For example, it is possible that increased internationalization will lead to greater

market learning and improved international vision. In this study, our goal was to provide a theoretically parsimonious model aimed at advancing extant perspectives. The nature and role of feedback loops is a possible area for future research. For the present, our instrumental variable analysis suggests that reverse causality is not likely to be a factor affecting study results. Our two-stage research design facilitated the discovery of theoretical relationships within a conceptual model that was then tested in the quantitative study. We sought to develop a testable theoretical framework comprised of key constructs that appear to exert the greatest impact on early internationalization.

Conclusion

We report an investigation of the enablers of early internationalization with a view to enhancing scholarly understanding of organizational learning and its role in firm performance. Moving beyond the emphasis of market learning within the strategic marketing and internationalization literatures, we address the need for a comprehensive explanation of the role of organizational learning on firm performance. We found the presence of an assortment of dynamic capabilities that includes learning capabilities aimed at creating multiple types of knowledge and a marketing capability, built and nurtured by visionary founders in their quest for international market entry. The organizational subsystems framework facilitates examining the complex interplay of these capabilities in their contribution to innovation and subsequently to early internationalization. Overall, our findings suggest that in spite of several decades of scholarly scrutiny, organizational learning remains a fertile ground for academic inquiry.

Table 1 Characteristics of case studies with early internationalization outcomes

Case	Interviews	Nature of business	Technical /Nontechnical	Export Intensity	Year of inception	Year first exported	Markets served (in order of entry)
A	Founders / CEO	Swim suits / online B2C	Nontechnical	90%	2002	2002	USA and Europe + 60 markets;
B	Founders/ CEOs	Bathing products B2C	Nontechnical	70%	2001	2003	USA, Asia, Australia
C	Founders / CEOs	Designer bags B2B/B2C	Nontechnical	60%	1997	1998	Europe, USA, Asia
D	Founders/ CEOs	Security surveillance B2B	Technical	60%	2001	2004	USA, Australia, Europe
E	Founder/ CEO	Pollution control B2B	Technical	80%	1999	2003	USA, Europe
F	Founders/ CEOs	Laser gun facilitated role playing game / B2B /B2C	Tech / non-tech	70%	1991	1994	USA
G	Founders/ CEO	Electrical switchgear B2B	Technical	60%	2002	2005	70 countries including USA, UK, Latin America
H	Current CEO and Founder	Smelter technology B2B	Technical	70%	1993	1996 (consultancy)	Korea, China, South America
I	CEO and Founder / Inventor	Tissue culture B2B	Technical	80%	1994	1995	USA, China, Canada and India
J	CEO and founding team	Biometric identification B2B	Technical	70%	2000	2002	Singapore, UK and Japan
K	CEO and founding team	Biodegradable plastics B2B	Technical	80%	2002	2004	Germany
L	CEO, CFO and Founding team	Transdermal delivery of medicine B2B	Technical	90%	1998	2000	USA,
M	CEO/ Founding team	Highly innovative shopping trolley B2B (no exporting within the 3 year cut off)	Tech / non-tech	0%	2002	Nil	Nil
N	CEO / Founder and Marketing Manager	Auto safety product B2B (no exporting within the 3 year cut off)	Technical	0%	2004	Nil	Nil

Table 2: Battery of items and item loadings for the two samples

	Item	Item loadings (Aus sample)	Item loadings (US sample)
International vision	The organizational culture (collective value system) was conducive to active exploration of new business opportunities abroad.	0.68	0.75
	Your firm continuously communicated the mission 'to succeed in international markets' to firm employees.	0.73	0.82
	Your firm developed human and other resources for achieving its goals in international markets.	0.75	0.75
	The top managers communicated information throughout the firm with respect to our successful and unsuccessful customer experiences abroad.	0.72	0.70
	The top managers were willing to go to great lengths to make our products or services succeed in foreign markets.	0.72	0.70
	The vision and drive of top managers were important in our decision to enter foreign markets.	0.79	0.64
Internally focused learning capability	Made substantial investments in equipment needed for research and experimental activities.	0.67	0.65
	Made substantial investments in staff involved with research and experimental activities.	0.68	0.69
	Identified the significance of intellectual property in building its technological advantage.	0.65	0.61
	Used new knowledge from research and experimental activities to resolve technological problems.	0.86	0.79
	Used new knowledge from research and experimental activities to improve the enterprise.	0.86	0.76
	Recombined knowledge from research and experimental activities in new ways to develop innovative products and services.	0.87	0.77
	Transferred knowledge from research and experimental activities to new projects and activities.	0.87	0.70
Network Learning capability	Undertook extensive networking with external research institutions to acquire technical or non-technical knowledge.	0.68	0.80
	Acquired technical or non-technical knowledge through attendance at industry gatherings and international conferences.	0.65	0.78
	Combined new knowledge generated through networks with existing technical or non-technical knowledge.	0.80	0.65
	Used new knowledge gained through networks to resolve customer focused problems.	0.80	0.76
	Transferred knowledge generated through networks to new projects and activities.	0.82	0.84
	Redeployed the resources in unproductive network learning projects to productive ones.	0.80	0.79

Marketing capability	Invested in developing products or services having a global appeal.	0.55	0.74
	Used promotional activities (e.g. advertising) of the firm to rapidly gain international market share/sales growth.	0.64	0.71
	Used its marketing resources and skills to effectively meet the value expectations of the targeted customers.	0.79	0.68
	Combined its marketing resources and skills more effectively to meet the customer needs.	0.87	0.69
	Recombined its marketing resources to meet changing customer expectations and competitor activity.	0.81	0.66
Market-focused learning capability	Gathered extensive information on customer needs.	0.66	0.65
	Proactively worked with lead customers to understand market needs.	0.70	0.69
	Combined new knowledge generated from customers and competitors with existing market knowledge.	0.77	0.81
	Used the new knowledge generated from markets to develop and refine the product or service concepts.	0.76	0.75
	Transferred knowledge generated from markets to new projects and activities.	0.79	0.67
	Transformed knowledge generated from markets and applied this to new projects and activities.	0.81	0.75
Innovation performance	Product or service innovations introduced by your firm were...	0.75	0.70
	Product or service innovations have been mainly...	0.72	0.71
	Process innovations introduced by your firm were...	0.72	0.67
	Process innovations have been mainly...	0.74	0.80
	Market establishment innovations introduced by your firm were...	0.73	0.68
	Market establishment innovations have been mainly...	0.74	0.75
Early internationalization	Successfully marketed to a chosen market abroad.	0.78	0.72
	Successfully entered multiple markets abroad.	0.66	0.77
	Entered markets abroad within a short period of time.	0.63	0.81
	Increased revenue from the firm's activities abroad.	0.82	0.78
	Increased market share in markets abroad.	0.83	0.73
	Increased customer satisfaction in markets abroad.	0.70	0.68
	Established a viable business due to activities abroad.	0.71	0.86

Table 3: Summary statistics: Australian sample

Construct		Items in scale	Mean	S.D.	AVE	Cronbach's alpha/CR	(a)	(b)	(c)	(d)	(e)	(f)	(g)
International vision	(a)	6	4.20	0.64	0.54	0.83/0.87	0.54						
Internally focused learning capability	(b)	7	3.59	0.90	0.61	0.89/0.92	0.21	0.61					
Market-focused learning capability	(c)	6	3.86	0.72	0.56	0.84/0.88	0.29	0.34	0.56				
Network learning capability	(d)	6	3.18	0.87	0.58	0.85/0.89	0.09	0.18	0.25	0.58			
Marketing capability	(e)	5	3.76	0.75	0.55	0.79/0.86	0.34	0.24	0.44	0.26	0.55		
Innovation performance	(f)	6	3.27	0.88	0.54	0.83/0.87	0.14	0.35	0.26	0.15	0.23	0.54	
Early internationalization	(g)	7	4.04	0.69	0.54	0.86/0.89	0.17	0.12	0.19	0.11	0.25	0.15	0.54

N=201. Values in the shaded diagonal represent the average variance extracted (AVE) for each construct. The squares of the correlation estimates are presented in the lower triangle of the matrix. The squared correlation estimate should be lower than the AVE for discriminant validity to be established between two constructs; S.D. – Standard deviation; CR – Composite reliability.

Table 4: Summary statistics: U.S. sample

Construct		Items in scale	Mean	S.D.	AVE	Cronbach's alpha/CR	(a)	(b)	(c)	(d)	(e)	(f)	(g)
International vision	(a)	6	3.85	0.43	0.53	0.82/0.87	0.53						
Internally focused learning capability	(b)	7	3.62	0.48	0.51	0.84/0.88	0.10	0.51					
Market-focused learning capability	(c)	6	3.69	0.43	0.52	0.82/0.87	0.34	0.36	0.52				
Network learning capability	(d)	6	3.59	0.49	0.60	0.86/0.90	0.08	0.49	0.41	0.60			
Marketing capability	(e)	5	3.87	0.40	0.49	0.74/0.82	0.43	0.12	0.38	0.11	0.49		
Innovation performance	(f)	6	3.35	0.52	0.52	0.82/0.87	0.25	0.07	0.30	0.17	0.25	0.52	
Early internationalization	(g)	7	4.09	0.50	0.59	0.88/0.91	0.33	0.01	0.26	0.02	0.29	0.29	0.59

N=234. Values in the shaded diagonal represent the average variance extracted (AVE) for each construct. The squares of the correlation estimates are presented in the lower triangle of the matrix. The squared correlation estimate should be lower than the AVE for discriminant validity to be established between two constructs; S.D. – Standard deviation; CR – Composite reliability.

Figure 1 The role of the market subsystem and the socio-technical subsystem in innovation and firm performance: Conceptual model

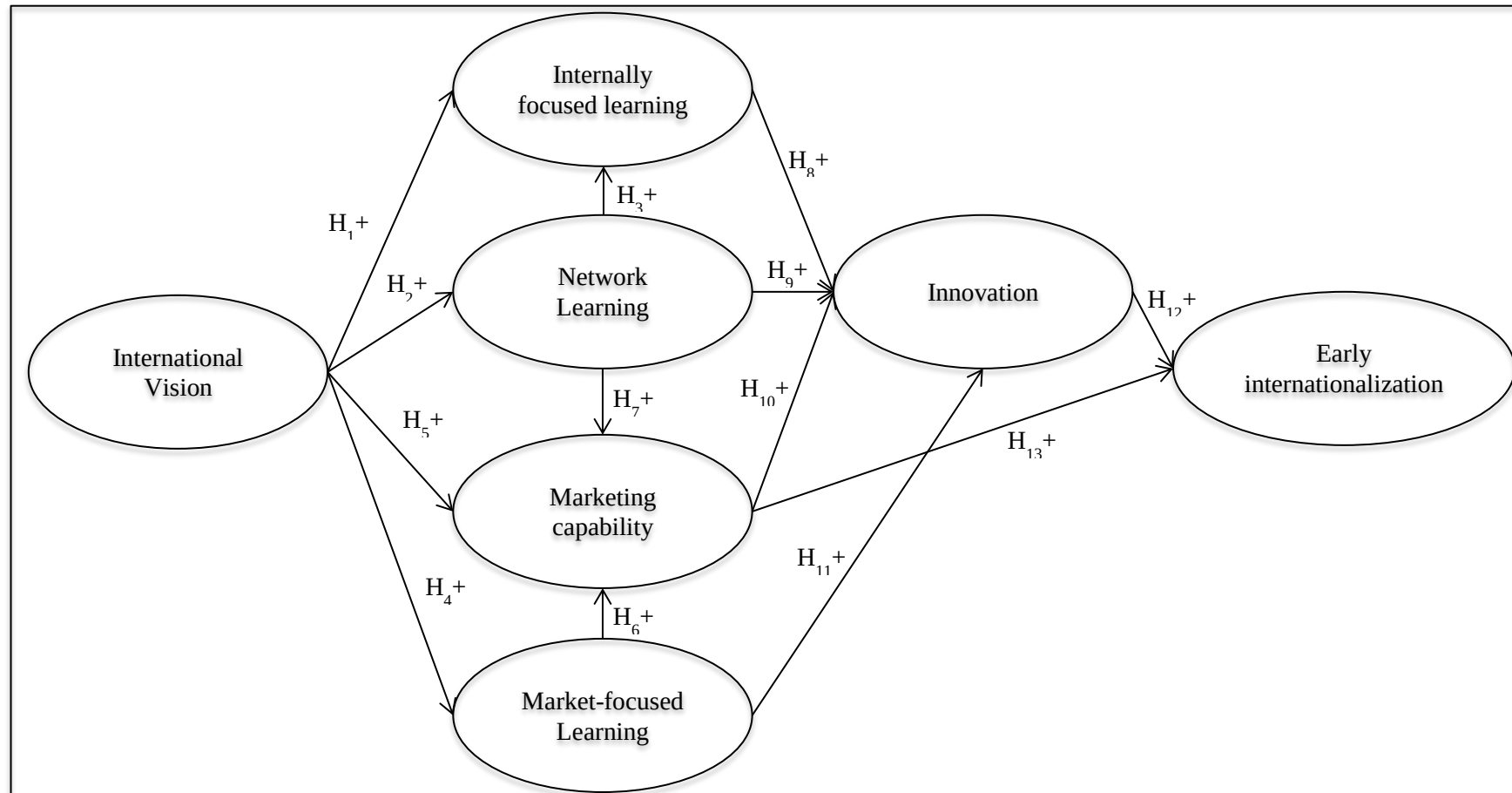


Figure 2: Structural model: Australian sample

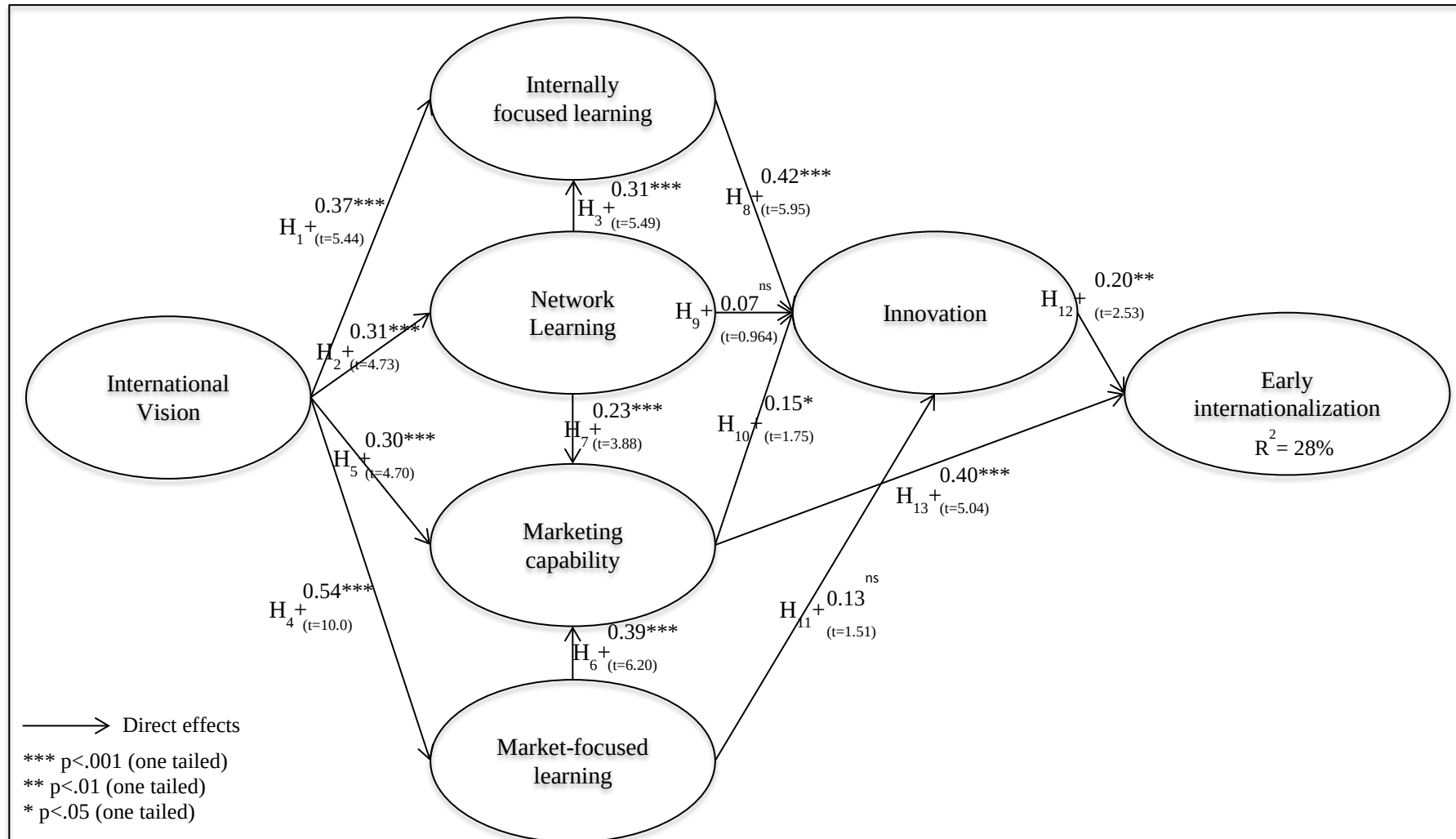
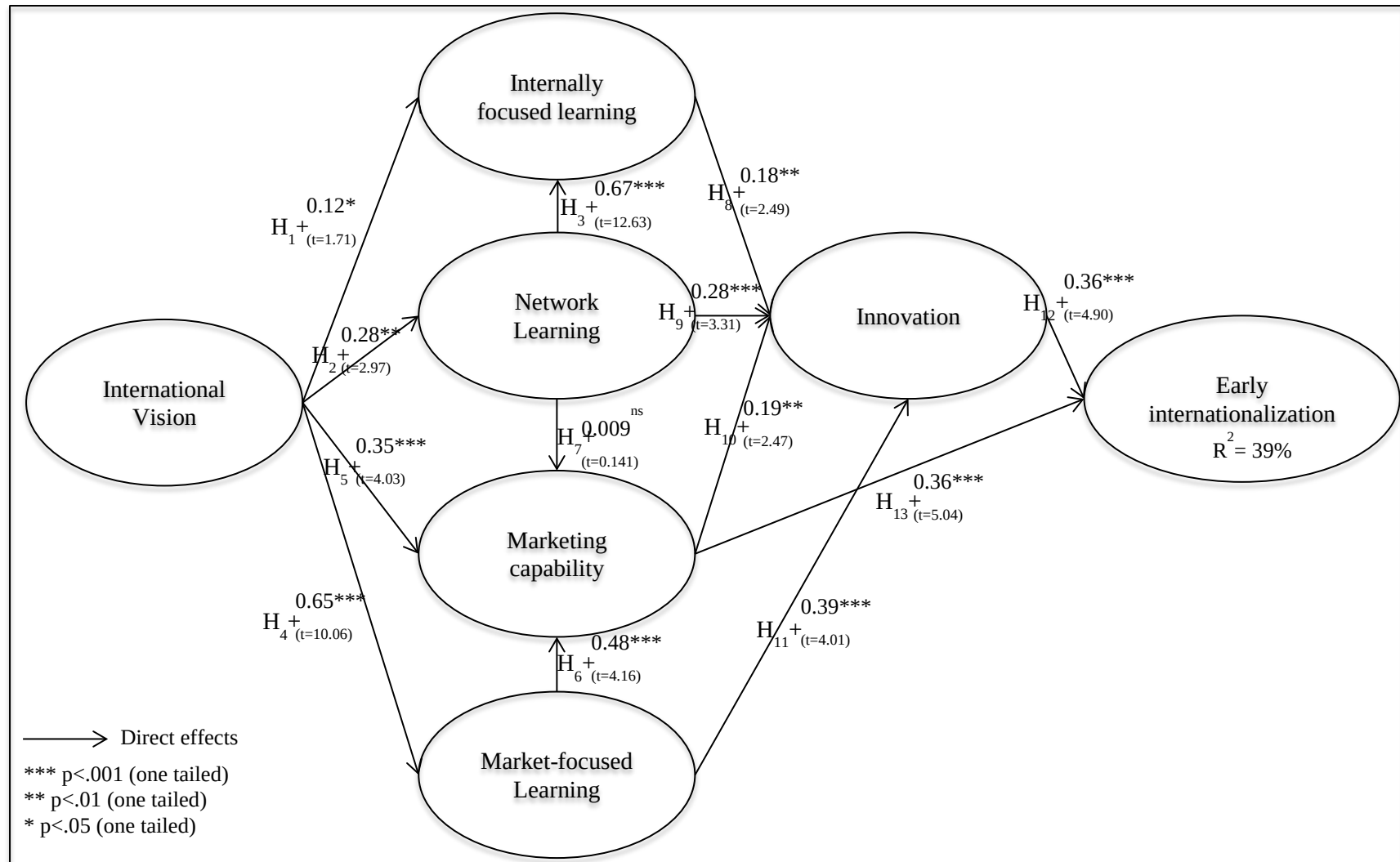


Figure 3: Structural model: U.S. sample



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